

DesignLab Tutorial

Getting started. The installation of DesignLab is easy. You just need to pick either a Windows or a Linux download from this month's DVD. The Windows download provides you with an easy setup.exe file, while the Linux 32-bit and 64-bit versions provide a tarball that contains a setup script. Additionally, to use DesignLab, you will also need to install the free Xilinx ISE WebPACK software.

Quick Tutorial. If this is your first time with DesignLab IDE, and you are not sure where to start, then this tutorial is for you. In this example we will see how to load an FPGA circuit and corresponding sketch to your Papilio board.

Open QuickStart sketch. Once you have opened the DesignLab IDE, you will find yourself in a table of contents page. Scroll down to find the Papilio DUO Quickstart sketch or the Papilio Pro/One Quickstart sketch.

Select the board type and serial port. It is important to select the board type and serial port that matches the Papilio board you are using. For the Papilio One and Papilio Pro, you will want to select the second COM port that shows up.

<Select Board Type.png>

<Select Port Type.png>

Load circuit to FPGA. DesignLab adds an FPGA circuit to every sketch. In this step you want to load the circuit to the FPGA. There are three icons in the DesignLab editor window for working with circuits. Use these icons to

Load circuit icon:

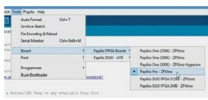


Fig. 1: Select Board Type

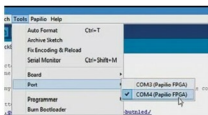


Fig. 2: Select Port Type



Fig. 3: Load-Circuit

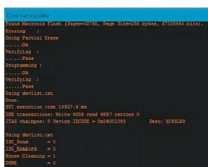


Fig. 4: Verifying-Loading-Circuit



Fig. 5: View-Circuit

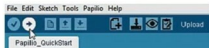


Fig. 6: Upload-Sketch



Fig. 7: Verify-Running

<Load-Circuit.png>

Verify the circuit is loaded:

<Verifying-Loading-Circuit.png>

View the circuit and make necessary edits:

<View-Circuit.png>

Upload sketch:

<Upload-Sketch.png>

After the FPGA circuit is loaded you can load a sketch to it if it contains a ZPUino soft processor.

Verify the sketch is running. The Quickstart sketch used in this tutorial outputs an ASCII table over the serial port, so we can open the serial monitor to verify that the sketch is running. Click on the serial monitor icon and you are good to go:

<Verify-Running.png>

system on chip (SoC) will satiate your thirst. DesignLab includes the ZPUino soft processor with a Wishbone bus, providing greater speed and flexibility than an Arduino-compatible chip. Making your own soft processor with just the right mix of peripheral circuits is known as an SoC design. DesignLab lets you build your own personalised SoC at the click of a few buttons.

DesignLab creates SoCs with ten serial ports, or a PWM on every pin, or something exotic like classic Atari and Commodore audio chips connected at the same time. The sky is the limit; you can create things that

don't exist anywhere else. (There could be a reason for that but hey, don't kill the messenger.)

Papilio circuit library. All the above is pretty useless if there isn't a supporting library of circuits (also known as cores) to do the heavy work. The IDE aims to provide the framework for anyone to write a core that can be wired into a circuit. And it does that very well.

Itching to get started?

Don't waste time. Start dragging and dropping those circuit libraries to build your first creative FPGA project.

The best way to experience FPGA design is to pick up a board and just go crazy with it. The Papilio boards have extensive documentation online, both from the company as well as from the hordes of users, which should suffice for a beginner. The whole idea behind Papilio is to decentralise FPGA design and remove the stigma of it being a difficult thing to achieve. The developers want you to get comfortable with using an FPGA, and this IDE aims to do just that. **RFV**

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